

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

across first base resistance R1 in volts is given by:

$$V_1 = V_{cc} - V_2$$

Therefore first base resistance in kilo-ohms is given by:

$$R_1 = (V_1 / V_2) \times R_2$$

Power consumed by the circuit in milliwatts is given by:

$$P = I_c \times V_{cc}$$

Collector voltage of the transistor in volts is given by:

$$V_c = I_c \times R_c$$

Voltage across the collector and emitter of the transistor in volts is given by:

$$V_{ce} = V_{cc} - (V_c + V_e)$$

Current flowing through resistor R1 in milliamperes is given by:

$$I_1 = V_1 / R_1$$

Current flowing through resistor R2 in milliamperes is given by:

$$I_2 = V_2 / R_2$$

Power dissipated across resistor R1 in milliwatts is given by:

$$PR_1 = I_1 \times I_1 \times R_1$$

Power dissipated across resistor R2 in milliwatts is given by:

$$PR_2 = I_2 \times I_2 \times R_2$$

Power dissipated across collector resistor R_c in milliwatts is given by:

$$PR_c = I_c \times I_c \times R_c$$

Power dissipated across emitter resistor R_e in milliwatts is given by:

$$PR_e = I_e \times I_e \times R_e$$

Coupling capacitors C1 and C2 and bypass capacitor C_e in microfarads is given by:

$$C = C_1 = C_2 = C_e = (1000000) / (2 \times 3.14 \times f \times X_c)$$

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Enter the value of supply voltage Vcc in volts:9
Enter the value of collector current Ic in milliamper:2
Enter the value of current gain of the transistor b:200
Enter the value of Minimum frequency to be amplified f in Hz:20
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Fig. 2: Input parameters for the example entered by user

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Voltage across R1 as V1=7.400000 volts
Voltage across R2 as V2=1.600000 volts
Power consumed P=10.000000 milli watts

Value of resistance R1=41.625000 kilo ohms
Value of resistance R2=9.000000 kilo ohms
Value of resistance Rc=1.800000 kilo ohms
Value of resistance Re=0.450000 kilo ohms
Input & output coupling capacitors C1=C2=7.961783 micro farad
Value of bypass capacitor Ce=7.961783 micro farad
Collector voltage of the transistor Vc=3.600000 Volts
Emitter voltage of the transistor Ve=0.900000 Volts
Voltage across colr.and emitter of the transistor Vce=4.500000 Volts
Base current of the transistor Ib=0.010000 milliamper
Current flowing through resistance R1 is I1=0.177778 milliamper
Current flowing through resistance R2 is I2=0.177778 milliamper

Power dissipated across the resistance R1 is PR1=1.315556 milli watts
Power dissipated across the resistance R2 is PR2=0.284444 milli watts
Power dissipated across the resistance Rc is PRc=7.200000 milli watts
Power dissipated across the resistance Re is PRRe=1.818045 milli watts
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Fig. 3: Screenshot of an example program output

where f is the minimum frequency in hertz to be amplified by the transistor amplifier and X_c is the reactance of capacitors C1, C2 and C_e.

Reactances of these three capacitors are considered to be the same, which is 1000 ohms.

Software program

All 19 formulae mentioned above are programmed using C++ language. The circuit has been tested using Turbo C++ version 3 and Visual C++ IDE on Windows 7.

After running the program, it prompts you to enter the values of DC supply voltage V_{cc}, bias collector current of transistor I_c, current gain of transistor b and minimum frequency f to be amplified by the transistor amplifier.

For example, enter V_{cc} = 9, I_c = 2, b = 200 and f = 20 as shown in Fig. 2.

Next, Hit Enter. Within a fraction of a second, all 19 parameters for the transistor amplifier are calculated and displayed on the monitor as shown in Fig. 3.

It is found that, to calculate the above 19 parameters, around 12 minutes are required. Therefore in order to save time, such a program can be quite useful for a circuit designer. ●



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